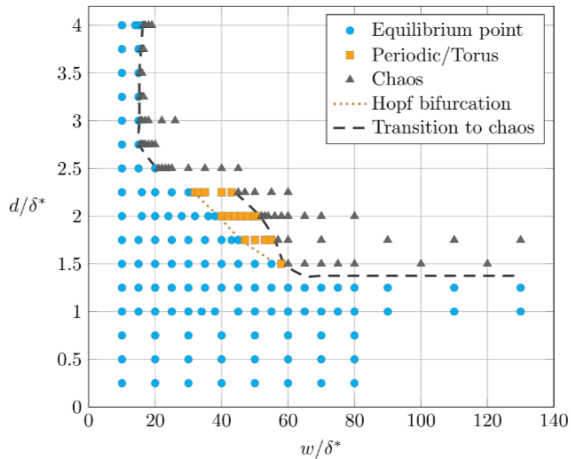


# IMPERIAL

## Several things

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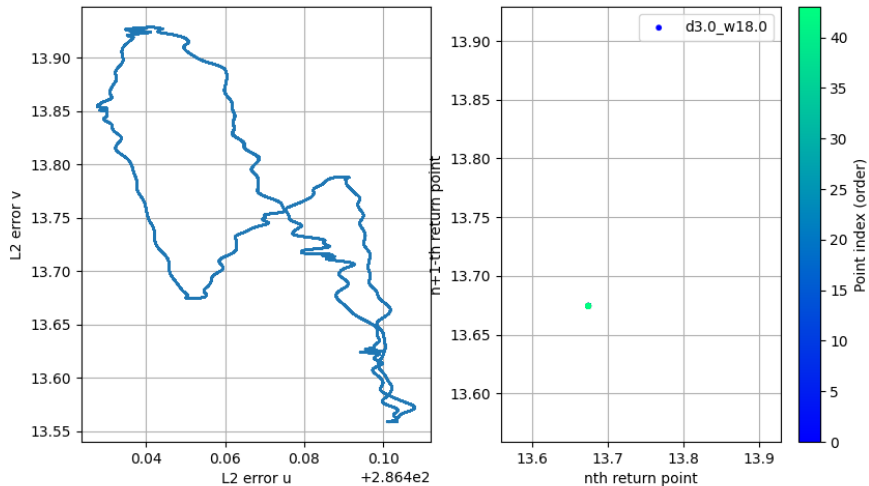
# Reminder



## Homework

- Interpolating the solution at  $d = 3$  and  $w = 15.5$  (not equilibria) into the geometry of  $d = 3$  and  $w = 15$  and running the simulation, the solution converges to the same equilibrium point.
- I have the feeling that the attractor that we were observing starting around 15.5 it might be another Hopf bifurcation.

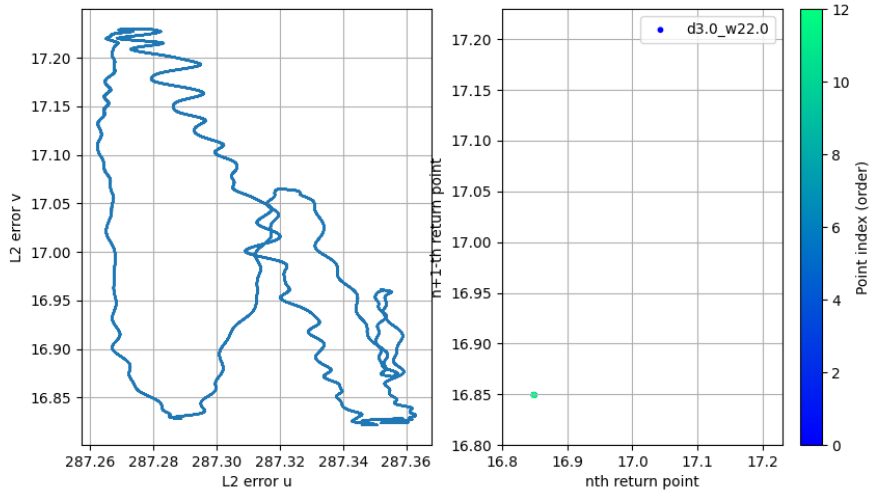
## $d = 3$ and $w = 18$ return map



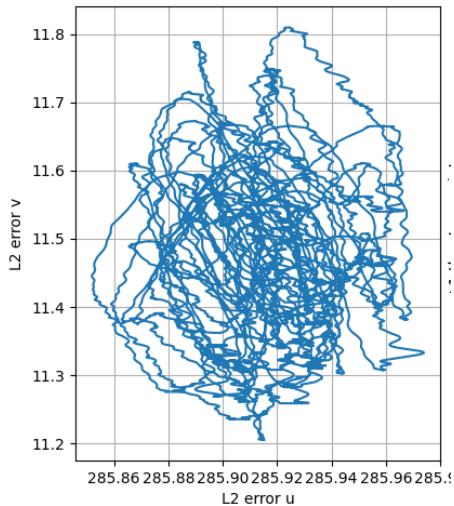
$d = 3$  and  $w = 18$  return map



## $d = 3$ and $w = 22$ return map



**But  $d = 3$  and  $w = 16$  doesn't exhibit this behaviour... Why?**



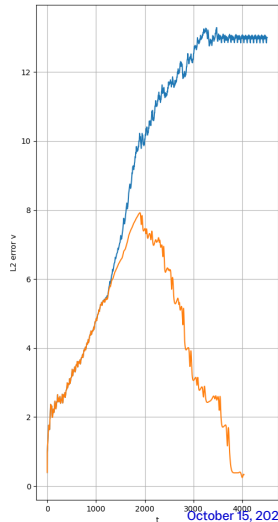
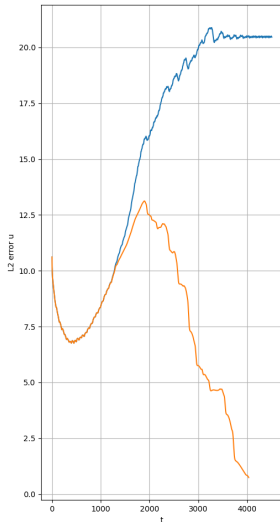
**But  $d = 3$  and  $w = 16$  doesn't exhibit this behaviour... Why?**



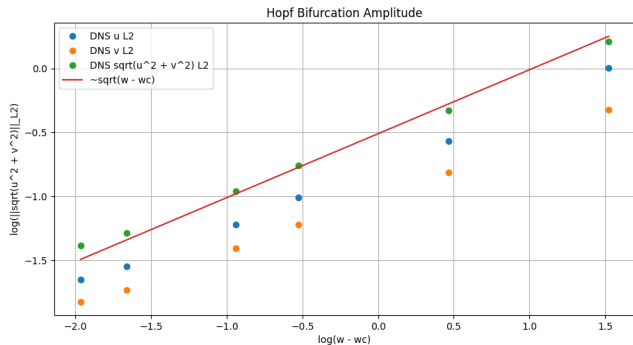
- Looks very similar to the previous one at  $w = 18$ .
- Numerical issue (e.g. SVV, etc) or physical?

## Edge state at $d = 3$ and $w = 17$

- We choose the L2 norm in the domain as a measure of the amplitude of the perturbation.
- Difference between initial conditions:  
 $||\mathbf{u}_{\text{div}}||_2 - ||\mathbf{u}_{\text{conv}}||_2 \sim 4 \cdot 10^{-9}$   
 $||\mathbf{v}_{\text{div}}||_2 - ||\mathbf{v}_{\text{conv}}||_2 \sim 1 \cdot 10^{-10}$
- I have the feeling that the unstable state, is just pumping vortices “differently” from the stable one.



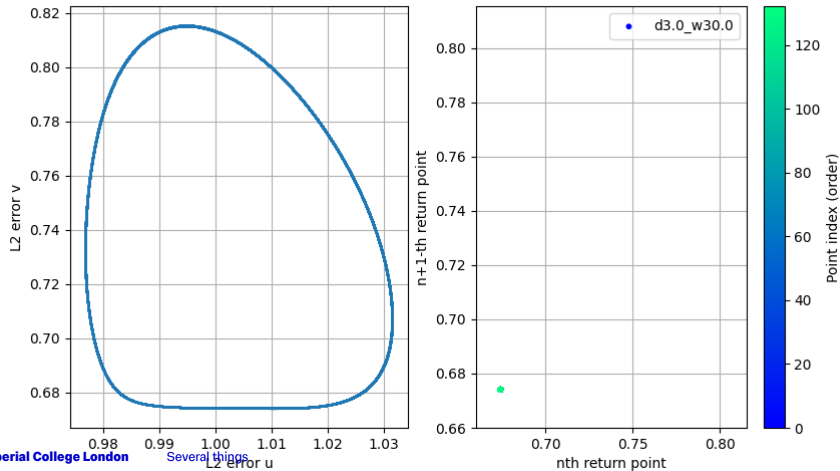
# Hopf bifurcation amplitude



- Any amplitude needs to follow  $A(w) \sim \sqrt{w - w_c}$
- $w_c \in (25.41, 25.42)$
- Data from  $w \in \{25.55, 25.6, 25.8, 26, 27, 30\}$ :
- At  $w = 35$  we already have a torus.

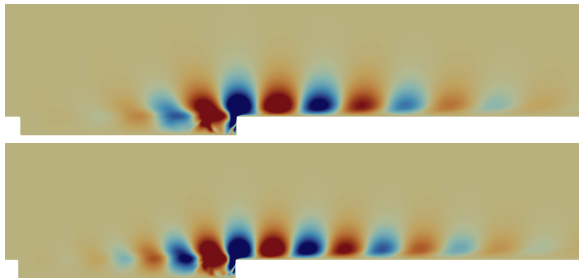
## Edge state at $d = 3$ and $w = 30$

- Periodic Orbit, projected to the new system  $\|u\|_2$  vs  $\|v\|_2$ .



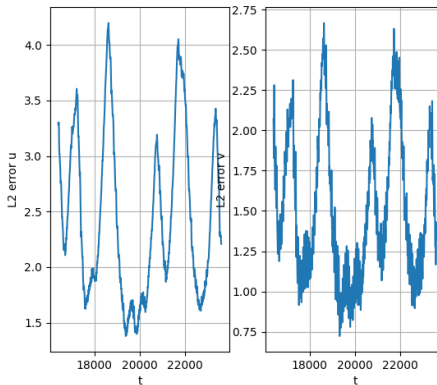
## Edge state at $d = 3$ and $w = 35$

- TORUSSSSS!!!!
  - From Global stability analysis, we found two unstable modes with different frequencies:
    - $\sigma = 0.012835, \omega = 0.21905$
    - $\sigma = 0.005983, \omega = 0.26979$
- Both modes look very very similar:
- Reminder. TS waves freq around the gap:  $\sim 0.08$



## Edge state at $d = 3$ and $w = 35$ return map?

- Energy doesn't have any clear distinguishable shape. Why? I think because the two frequencies of the unstable modes couple and propagate the energy to other frequencies exciting the TS modes. This is NOT a transient phenomenon, it is always like this.



## Edge state at $d = 3$ and $w = 35$ return map?

- Energy doesn't have any clear distinguishable shape. Why? I think because the two frequencies of the unstable modes couple and propagate to other energies exciting the TS modes. This is NOT a transient phenomenon, it is always like this.



## Edge state at $d = 3$ and $w = 35$ return map?

- If we try to do the return map ( $u$  vs  $v$ ) very locally in a point just above the gap, we see:
- Of course the shape of the scatter plot in the return maps depends on the point we choose.

